

MH1101 Tutorial 1 (Week 2)

Reference: Sections 1.1, 1.2, 1.3 (Lecture Notes)

1.

- (i) Find the most general antiderivative of the function $\frac{5 - 4x^3 + 2x^6}{x^6}$, $x \in \mathbb{R} \setminus \{0\}$.
- (ii) Find f if $f'(x) = \sec x(\sec x + \tan x)$, $-\pi/2 < x < \pi/2$, $f(\pi/4) = -1$.
- (iii) Find f if $f''(x) = \sin x + \cos x$, $x \in \mathbb{R}$, and $f(0) = 3$, $f'(0) = 12$.

2. Using the right endpoints as sample points, find the Riemann sum that corresponds to the definite integral $\int_1^4 (x^2 + 2x - 5) dx$.

Evaluate the integral by computing the limit of the Riemann sum directly. You may assume that $\sum_{i=1}^n i = \frac{n(n+1)}{2}$, $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$.

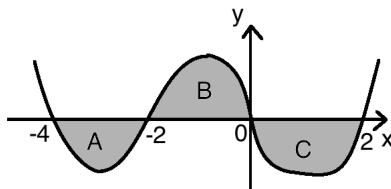
3. Evaluate the integral by interpreting it in terms of areas:

- (i) $\int_0^9 \left(\frac{1}{3}x - 2 \right) dx$.
- (ii) $\int_{-3}^0 \left(1 + \sqrt{9 - x^2} \right) dx$.

4. Use the properties of integrals to verify the inequality without evaluating the integrals.

- (i) $2 \leq \int_{-1}^1 \sqrt{1 + x^2} dx \leq 2\sqrt{2}$.
- (ii) $\int_0^{\pi/2} x \sin x dx \leq \frac{\pi^2}{8}$.

5. Each of the regions A , B , and C bounded by the graph f and the x -axis has area 3.



Find the value of $\int_{-4}^2 (f(x) + 2x + 5) dx$ by interpreting definite integrals as areas (net areas).

6. Express the following limit as a definite integral:

(i) $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{i^4}{n^5}.$

(ii) $\lim_{n \rightarrow \infty} \frac{2}{n} \sum_{i=1}^n \frac{1}{1 + (1 + 2i/n)^2}.$

7. If f is a continuous function on $[a, b]$, show that

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx.$$

8. Suppose $f(x)$ is linear on $[a, b]$ (i.e. $f(x) = mx + c$ for some constants m and c), and $f(x) > 0$ on $[a, b]$. Explain, using a graph, that

$$\int_a^b f(x) dx = \frac{1}{2}(R_n + L_n) \text{ for all } n,$$

where R_n and L_n are right-endpoints and left-endpoints approximations of the area under f and above $[a, b]$.

Answer Keys.

1. (ii) $f(x) = \tan x + \sec x - 2 - \sqrt{2}$ (iii) $f(x) = -\sin x - \cos x + 13x + 4$
2. 21 3. (i) $-\frac{9}{2}$ (ii) $3 + \frac{9}{4}\pi$. 5. 15